The German contribution to the European Bioinformatiomatics Research Infrastructure for Life Sciences ELIXIR: Services, Training & Cloud Computing

Rainer Schwacke[1], Marie Bolger[1], Jan P. Buchmann[2], Daniel Arend[3], Heidrun Gundlach[4], Matthias Lange[3], Klaus F. X. Mayer[4], Uwe Scholz[3], Björn Usadel[1,2], and Sebastian Beier[1]

[1]Forschungszentrum Jülich GmbH, CEPLAS, BioSC, Institute of Bio- and Geosciences, Bioinformatics (IBG-4), 52425 Jülich, Germany
[2]Heinrich-Heine-University Düsseldorf, CEPLAS, BioSC, Institute of Biological Data Science, 40225 Düsseldorf, Germany
[3]Leibniz Institute of Plant Genetics and Crop Plant Research (IPK), 06466 Seeland, Germany
[4]Helmholtz Center Munich-German Research Center for Environmental Health, Plant Genome and Systems Biology, 85764 Neuherberg, Germany

The "German Network for Bioinformatics Infrastructure - de.NBI" has become an important player in international research through its contribution to ELIXIR. These resources include databases, software tools, training materials, cloud storage and supercomputers. In this context, de.NBI, as the German Node, makes important contributions to challenges such as research data management and supporting the European Open Science Cloud (EOSC) with the de.NBI cloud infrastructure.

Within the framework of de.NBI, partners from IPK Gatersleben, HMGU Munich and Forschungszentrum Jülich provide plant bioinformatics.

The main focus of de.NBI / ELIXIR-DE here is on the software tools and services offered. These are among others: e!DAL-PGP [1], a repository for plant genomics and phenomics data; MISA-Web [2] (part of the Crop Analysis Tools Suite), a tool for searching microsatellites in genomic sequences; Mercator4 [3], a tool for classification of plant protein sequences; Trimmomatic [4], a tool for trimming NGS data; as well as a tailored consulting service for genome and repeat annotation.

In addition to the tools, training courses and summer schools are also offered on a regular basis. These cover everything from learning how to program, to the correct use of the offered tools, and the construction of complex workflows with a focus also on research data management.

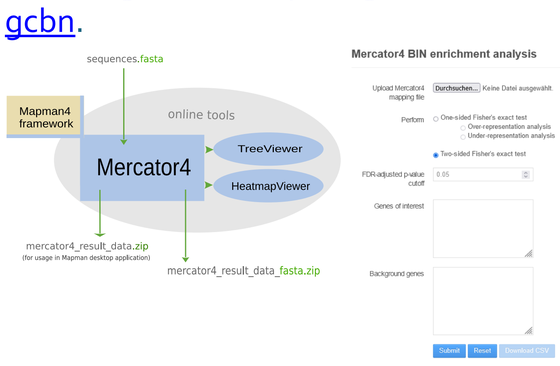
With this portfolio, there are also many points of connection to various German NFDI initiatives (e.g. DataPLANT, FAIRagro and NFDI4Biodiversity) and non-European communities (e.g. AgBioData and Australian BioCommons).

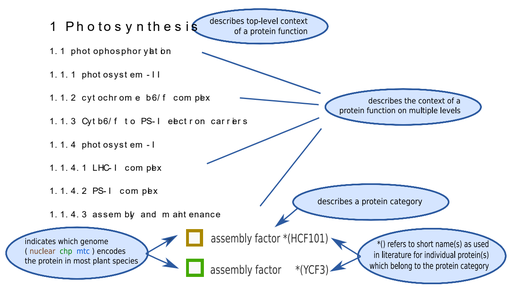
Services

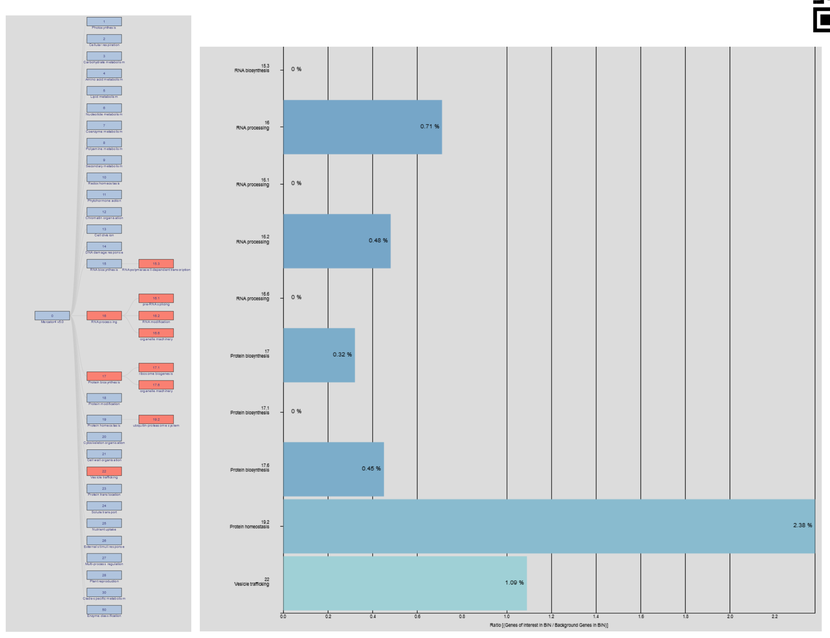
The service portfolio of the plant community within de.NBI consists of ten well-known and established tools and databases. These can be accessed through https://www.denbi.de/services-according-to-service-centers/german-crop-biogreenformatics-network-gcbn.

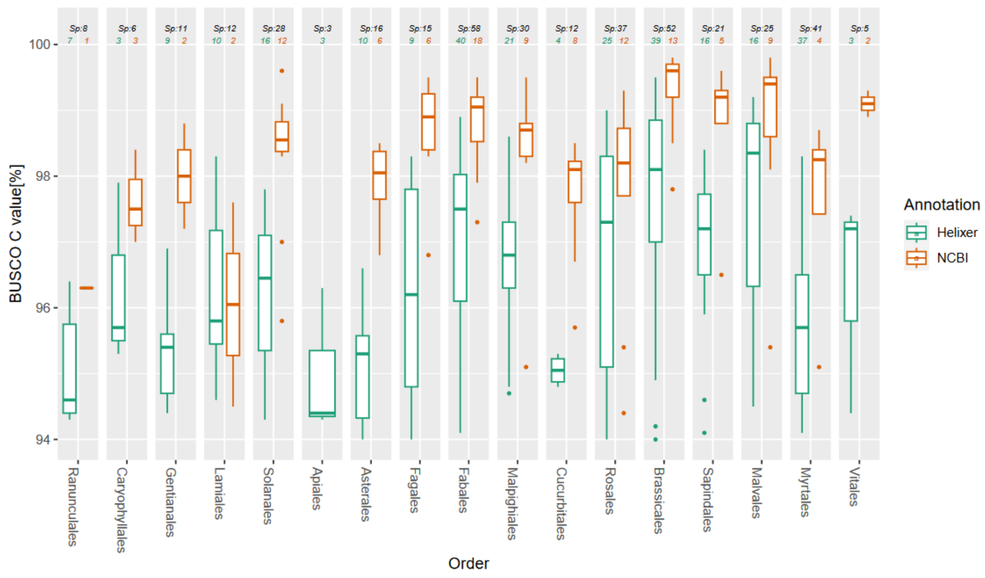
Services are maintained and permanently improved. For this purpose, user feedback surveys and KPIs are collected and evaluated on a monthly basis. An exhaustive list of KPIs is available under: https://doi.org/10.5281/zenodo.6597826

New services are regularly added to the existing services when needed. Currently, the novel deep learning based tool for the generation of structural gene predictions Helixer [5] is being evaluated. Initial tests showed similar BUSCO values like reference-quality gene annotations, outperforming many established tools and workflows while reducing memory and computational time requirements.









Training

de.NBI offers a variety of training formats, ranging from traditional or virtual workshops to summer schools and practical demonstrations. Since 2015, GCBN has assisted in 22 training events with 352 participants. Additionally, "train-the-trainer" courses have been conducted in recent years and knowledge sharing between ELIXIR nodes has been facilitated through the Staff Exchange program.

de.NBI Spring School [2023] Data management

BioHackathon Germany 12.-16.12.2022

BioHackathon Germany 11.-15.12.2023

de.NBI Summer School [2023] Bio Data Analysis for Life Sciences

Cloud Computing

Users

Registered: 2401

Projects

Cumulative Projects: 791

OpenStack - Cumulative: 491

SimpleVM - Cumulative: 300

Currently active Projects: 521

OpenStack - currently active: 348

SimpleVM - currently active: 173

Resources

Cores:

78300 cores allocated

66973 cores allocated by OpenStack projects

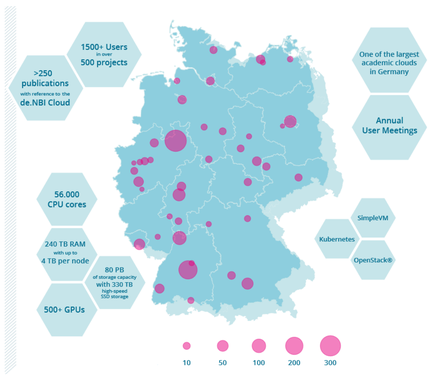
11327 cores allocated by SimpleVM projects

RAM:

460068 GB of RAM allocated

396543 GB allocated by OpenStack projects

63525 GB allocated by SimpleVM projects



Cloud computing is provided by de.NBI free of charge to the academic community in the form of the de.NBI Cloud. Both storage and computing resources are operated in a cloud federation at six German cloud centres. Access to the de.NBI Cloud can be requested by Principal Investigators of any German University or Research Institution through a web form on the de.NBI Cloud Portal (https://cloud.denbi.de/).

Contact

@basti_beier
@ELIXIREurope

/in/beiersebastian
/company/elixir-europe

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